

Extra queries on Supplier database

1. Find the most expensive part overall and the supplier who supplies it.
2. Find suppliers who do NOT supply any red parts.
3. Show each supplier and total value of all parts they supply.
4. Find suppliers who supply at least 2 parts cheaper than ₹20.
5. List suppliers who offer the cheapest cost for each part.
6. Create a view showing suppliers and the total number of parts they supply.
7. Create a view of the most expensive supplier for each part.
8. Create a Trigger to prevent inserting a Catalog cost below 1.
9. Create a trigger to set to default cost if not provided.

ANSWERS: -

```
1 • create database sup;
2 • use sup;
3
4 • CREATE TABLE Supplier (
5     sid INT PRIMARY KEY,
6     sname VARCHAR(50),
7     address VARCHAR(100)
8 );
9
10 • CREATE TABLE Parts (
11     pid INT PRIMARY KEY,
12     pname VARCHAR(50),
13     color VARCHAR(20)
14 );
15
16 • CREATE TABLE Catalog (
17     sid INT,
18     pid INT,
19     cost INT,
20     PRIMARY KEY (sid, pid),
21     FOREIGN KEY (sid) REFERENCES Supplier(sid),
22     FOREIGN KEY (pid) REFERENCES Parts(pid)
23 );
24
```

```

25 • INSERT INTO Supplier VALUES
26     (10001, 'Acme Widget', 'Bangalore'),
27     (10002, 'Johns', 'Mysore'),
28     (10003, 'Reliance', 'Chennai'),
29     (10004, 'Tata', 'Mumbai');
30
31 • INSERT INTO Parts VALUES
32     (20001, 'Book', 'Red'),
33     (20002, 'Charger', 'Green'),
34     (20003, 'Mobile', 'Black'),
35     (20004, 'Pen', 'Red'),
36     (20005, 'Pencil', 'Blue');
37
38 • INSERT INTO Catalog VALUES
39     (10001, 20001, 50),
40     (10001, 20002, 60),
41     (10001, 20003, 150),
42     (10001, 20004, 35),
43     (10002, 20001, 55),
44     (10002, 20004, 45),
45     (10003, 20003, 120),
46     (10003, 20005, 40),
47     (10004, 20001, 65);
48
109 • SELECT p.pname, s.sname, c.cost
110 FROM Catalog c
111 JOIN Parts p ON c.pid = p.pid
112 JOIN Supplier s ON c.sid = s.sid
113 WHERE c.cost = (
114     SELECT MAX(cost)
115     FROM Catalog
116 );
117

```

Result Grid			
Filter Rows:			
	pname	sname	cost
▶	Mobile	Acme Widget	150

```

118 • SELECT s.sname
119 FROM Supplier s
120 WHERE NOT EXISTS (
121     SELECT *
122     FROM Parts p
123     JOIN Catalog c ON p.pid = c.pid
124     WHERE p.color = 'Red'
125     AND c.sid = s.sid
126 );
127

```

Result Grid | Filter Rows: | Export:

sname
Reliance

```

128 • SELECT s.sname, SUM(c.cost) AS total_value
129 FROM Supplier s
130 JOIN Catalog c ON s.sid = c.sid
131 GROUP BY s.sname;
132
133

```

Result Grid | Filter Rows: | Export:

sname	total_value
Acme Widget	295
Johns	100
Reliance	160
Tata	65

```

133 • SELECT s.sname
134 FROM Supplier s
135 WHERE s.sid IN (
136     SELECT c.sid
137     FROM Catalog c
138     WHERE c.cost < 20
139     GROUP BY c.sid
140     HAVING COUNT(c.pid) >= 2
141 );
142

```

Result Grid | Filter Rows: | Export:

sname

```

143 • SELECT p.pid, p.pname, s.sname, c.cost
144 FROM Parts p
145 JOIN Catalog c ON p.pid = c.pid
146 JOIN Supplier s ON s.sid = c.sid
147 WHERE c.cost = (
148     SELECT MIN(c2.cost)
149     FROM Catalog c2
150     WHERE c2.pid = p.pid
151 );
152

```

Result Grid				
Filter Rows:				
Export:				
	pid	pname	sname	cost
▶	20001	Book	Acme Widget	50
	20002	Charger	Acme Widget	60
	20004	Pen	Acme Widget	35
	20003	Mobile	Reliance	120
	20005	Pencil	Reliance	40

```

153 • CREATE VIEW SupplierPartCount AS
154 SELECT s.sid, s.sname, COUNT(c.pid) AS total_parts
155 FROM Supplier s
156 LEFT JOIN Catalog c ON s.sid = c.sid
157 GROUP BY s.sid, s.sname;

159 • CREATE VIEW MaxCostPerPart AS
160 SELECT p.pid, p.pname, s.sname, c.cost
161 FROM Parts p
162 JOIN Catalog c ON p.pid = c.pid
163 JOIN Supplier s ON s.sid = c.sid
164 WHERE c.cost = (
165     SELECT MAX(c2.cost)
166     FROM Catalog c2
167     WHERE c2.pid = p.pid
168 );

```

```
170     delimiter $$;
171 • CREATE TRIGGER prevent_low_cost
172     BEFORE INSERT ON Catalog
173     FOR EACH ROW
174     BEGIN
175     IF NEW.cost < 1 THEN
176         SIGNAL SQLSTATE '45000'
177         SET MESSAGE_TEXT = 'Cost cannot be less than 1';
178     END IF;
179     END $$;
180
181     delimiter $$;
182 • CREATE TRIGGER default_cost
183     BEFORE INSERT ON Catalog
184     FOR EACH ROW
185     BEGIN
186     IF NEW.cost IS NULL THEN
187         SET NEW.cost = 50;
188     END IF;
189     END $$;
```